

Parallel Consensus Routing: Training-Free Quality Control, Coordinated Hallucination, and Model-Pair Selection in Cost-Efficient LLM Inference

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Abstract—Cost-efficient large language model (LLM) inference is dominated by two paradigms: *pre-query routing*, which commits to a model before any output exists, and *sequential cascading*, which escalates cheap-to-expensive in series and accrues additive latency. We study Parallel Consensus Routing (PCR), a third paradigm that runs two inexpensive models simultaneously and uses their binary output agreement as a training-free quality oracle: agreed queries are resolved immediately, disagreements trigger exactly one frontier escalation. Extending an initial single-pair study (MMLU: 92.7% accuracy at 27.5% escalation with domain-calibrated thresholds), we run a balanced four-pair design — two within-family and two cross-family pairs, frontier and provider held constant — on a complete six-benchmark grid (ARC, OpenBookQA, CommonsenseQA, MMLU, TruthfulQA, MMLU-Pro; plus seven exploratory pairs and GSM8K), scoring eight routing policies on identical stored data. Five results emerge. (1) Coordinated hallucination (C-Hall)—two cheap models agreeing on a wrong answer—scales monotonically with question difficulty, not task type: from 2% on OpenBookQA to $\approx 26\%$ on MMLU-Pro, reaching mathematical-reasoning levels on a plain multiple-choice task. (2) A capability-matched cheap pair with PCR matches always-frontier accuracy on easy/medium questions at 40–70% lower cost, but on hard questions its accuracy is safe while its C-Hall (12–26%) is not. (3) Shared model lineage buys a large agreement premium on hard/adversarial benchmarks (+9 to +22 pp; CommonsenseQA, TruthfulQA, MMLU-Pro) and, suggestively, a coordinated-hallucination premium there too (TruthfulQA +6.6 pp, CI [+0.3, +12.8]); on easy/medium benchmarks neither effect appears. With two pairs per condition the C-Hall signal is a hypothesis for powered follow-up. What makes a pair usable at all is capability matching, not lineage. (4) A third cheap model (PCR-3, majority vote) reliably breaks ties but yields no accuracy or safety gain. (5) On free-tier APIs, two of {gpt-oss-20B, Qwen3.6-27B, Qwen3.8-27B} on a low-latency provider is the recommended default: solo accuracies within ~ 1 pp, ≈ 0.5 s latency, ≈ 5 –12% escalation. All experiments used free-tier inference and required \$0 marginal API spend.

Index Terms—LLM routing, cost-efficient inference, parallel consensus, coordinated hallucination, model-pair selection, capability matching, black-box API inference

I. INTRODUCTION

The LLM API marketplace is sharply tiered: frontier models command costs one to two orders of magnitude above small open models [1]. For throughput workloads, routing *all* queries to a frontier is prohibitively expensive and routing *all* to a cheap model forfeits quality on the hard tail. The engineering problem is: *given a query, which model or combination minimizes cost subject to a quality constraint?*

Three paradigms exist. **Pre-query routing** [2], [3] predicts the right tier from input features before any output is generated. **Sequential cascading** [4] queries cheap first and applies a learned scorer to decide escalation, adding one model’s latency per hop. **Self-verification routing** [7] lets a small model grade its own answer, but self-verification is noisy on reasoning [13].

We study a structurally different idea: **use two cheap models as quality oracles for each other**. Two independently deployed cheap models queried in parallel that converge on the same answer provide agreement evidence that is stronger than either model’s self-assessment. On disagreement, exactly one frontier call is made. The system is training-free, black-box compatible, and faster than a serial cascade on every agreed query. The analogy to speculative decoding [5], [6] is exact—draft-then-verify at the response level rather than the token level—but without the white-box probability access speculative decoding requires.

The failure mode that makes this non-trivial is **coordinated hallucination (C-Hall)**: two cheap models agreeing on the *same wrong* answer. C-Hall is invisible to the router—it is indistinguishable from correct consensus—so understanding when it occurs is a safety prerequisite, not an optimization.

a) Contributions: This paper (i) reproduces and extends an initial single-pair PCR study with a **balanced 4-pair design on a complete 6-benchmark grid** (frontier and provider held constant), plus seven exploratory pairs; (ii) shows **C-Hall scales with difficulty, not task type**, generalizing the earlier “knowledge vs. math” dichotomy to a difficulty-tier one; (iii) runs a **full 8-policy ablation** (Always-Cheap, Best-Cheap, Consensus, PCR-2, Oracle-2-cheap, Random, Always-Frontier, PCR-3) on identical data; (iv) characterizes the **model-family effect**—a large agreement premium and a suggestive coordinated-hallucination premium, both on hard/adversarial benchmarks only—and identifies **capability matching** as the constraint that makes a pair usable at all; (v) provides measured cost/latency and a concrete **practitioner pair-selection guide**.

II. RELATED WORK

A. Sequential Cascade Routing

FrugalGPT [4] defines an LLM cascade with a learned (DistilBERT) accept/escalate scorer, matching GPT-4 at up

to 98% cost reduction on classification, but requiring task-specific supervised training and incurring latency proportional to cascade depth. Hybrid LLM [3] predicts the small-vs-large quality gap pre-query (up to 40% fewer large calls, <1% quality drop) but cannot use observed output. AutoMix [7] is architecturally closest to PCR: a single model self-verifies before escalation, with a POMDP meta-verifier absorbing self-verification noise. Our cross-model agreement is stronger than self-verification in principle—independent models cannot share a self-delusion—though, as our matrix shows, shared training corpora still induce correlated errors.

B. Pre-Query Routing

RouteLLM [2] trains four router variants on Chatbot Arena preference data (>85% cost reduction on MT-Bench, 45% on MMLU). GraphRouter [25] models query-LLM relationships with GNNs; FORC [24] learns per-LLM score prediction via contrastive learning; RouterBench [21] standardizes evaluation. All require trained components; PCR requires none.

C. Multi-Model Consensus and Ensembles

Multi-agent debate [8] improves reasoning through iterative text exchange at $O(\text{rounds} \times \text{models})$ cost. Self-Consistency [10] majority-votes multiple samples from one model; LLM-Blender [9] fuses open-model outputs with a trained ranker; ChatEval [26] debates for evaluation. None exploit cost differentials between tiers, and none use a single-round binary agreement signal.

D. Speculative Decoding

Leviathan et al. [5] and Chen et al. [6] accelerate decoding with a draft model whose tokens the target verifies by rejection sampling ($2\text{--}3\times$ speedup, provable equivalence). This needs the target’s per-token distribution—unavailable for proprietary APIs. PCR realizes the same draft-then-verify separation at the *response* level on final text only.

E. Semantic Caching, Calibration, and Hallucination

vCache [11] shows global similarity thresholds carry no correctness guarantee and proposes per-embedding online learning; our per-domain/per-difficulty threshold table is the routing analogue. SelfCheckGPT [12] detects hallucination via multi-sample consistency; Kadavath et al. [13] show LLMs are poorly calibrated on multi-step reasoning. Coordinated hallucination—agreement *negatively* correlated with correctness on hard items—has not been characterized in the routing literature.

F. Benchmarks

MMLU [14], GSM8K [15], and ARC-Challenge [16] are standard. We add TruthfulQA [17] (adversarial misconceptions), MMLU-Pro [18] (10-option, reasoning-heavy), OpenBookQA [19], and CommonsenseQA [20]. Chain-of-thought [23] is suppressed to isolate the agreement signal.

G. Research Gap

No prior work evaluates a *parallel binary-consensus* quality oracle for cost reduction, characterizes coordinated hallucination and its difficulty-dependence, controls for model-family relatedness while holding capability constant, or provides an empirical pair-selection guide for practitioners. This paper does all four.

III. METHODOLOGY

A. Problem Formulation

Let M_A, M_B be cheap models and M_F a frontier model, with $\text{cost}(M_F) \gg \text{cost}(M_A) \approx \text{cost}(M_B)$. For a multiple-choice query q each model emits a letter $r \in \{A, \dots\}$. The **PCR-2** decision rule is

$$\text{output}(q) = \begin{cases} r_A(q) & r_A(q) = r_B(q) \quad [\text{agreement}] \\ M_F(q) & r_A(q) \neq r_B(q) \quad [\text{disagreement}] \end{cases} \quad (1)$$

with expected per-query cost $\mathbb{E}[\text{cost}] = \text{cost}(M_A) + \text{cost}(M_B) + \text{cost}(M_F) \cdot P(\text{disagree})$.

B. PCR-3: Majority Vote

PCR-3 runs three cheap models M_A, M_B, M_C ; if any two agree, their answer is taken; only if all three differ is M_F called. This trades one extra cheap call for a lower escalation rate and a majority-vote margin.

C. Policy Space

Every policy below is scored on the *same* stored per-question data (cheap answers, agreement, frontier answer on disagreements, per-call token and latency instrumentation), so no policy requires additional API calls except Always-Frontier (see §IV): *Always-Cheap-A/B* (one small model); *Best-Cheap* (the stronger of the two); *Consensus, no escalation* (trust agreement, fall back to M_A on disagreement—provably equal to Always-Cheap-A); *PCR-2*; *Oracle-2-cheap* (correct whenever *either* cheap model is correct—the ceiling for any two-cheap method); *Random* (coin flip M_A vs. M_F); *Always-Frontier*; *PCR-3*.

D. Agreement Signal and Outcome Categories

We optionally compute embedding cosine similarity $s \in [-1, 1]$ between raw responses (all-MiniLM-L6-v2 [22]). Three outcome categories: **correct consensus** ($r_A = r_B = \text{gold}$); **coordinated hallucination** ($r_A = r_B \neq \text{gold}$); **disagreement** ($r_A \neq r_B$).

E. Coordinated Hallucination Hypothesis

We hypothesize C-Hall arises from shared inductive biases in overlapping training corpora: on retrieval both models recall the same fact (beneficial), on procedural or adversarial items both may pattern-match the same wrong shortcut or misconception. We test two moderators: (a) benchmark **difficulty**, and (b) whether the pair is **within-family** (same developer/lineage) or **cross-family**.

TABLE I
LITERATURE REVIEW — RELATED WORK AND DIFFERENTIATION OF CONTRIBUTION

Reference	Year/Venue	Approach	Key Limitation	PCR Advantage
FrugalGPT [4]	2023 / TMLR	Sequential cascade, trained quality scorer	Labeled training data; serial latency	Training-free; parallel; bounded at one frontier call
Hybrid LLM [3]	2024 / ICLR	Pre-query difficulty-gap router	No output signal; pre-commit	Post-query observed agreement
RouteLLM [2]	2024 / arXiv	Preference-trained router (4 variants)	Human preference labels	Zero labeled data; any API pair
AutoMix [7]	2024 / ACL	Self-verify + POMDP router	Self-verification noisy; POMDP training	Cross-model verification; training-free
Multi-agent Debate [8]	2024 / ICML	Iterative multi-round debate	$O(\text{rounds} \times \text{models})$; high latency	Single-round; throughput-optimized
Self-Consistency [10]	2023 / ICLR	Majority vote, one model	Same tier; no cost cut	Cross-tier; cheap replaces expensive samples
Spec. Decoding [5], [6]	2023 / ICML	Draft-then-verify on token probs	White-box only	Black-box; final text only
vCache [11]	2025 / arXiv	Online per-embedding threshold	Caching, not routing	Per-domain/difficulty calibration for routing
SelfCheckGPT [12]	2023 / EMNLP	Multi-sample consistency	Consistency assumed positive	C-Hall: consistency as <i>negative</i> indicator on hard items
RouterBench [21]	2024 / arXiv	Routing benchmark	No algorithm proposed	Results compatible with its methodology

F. Capability Matching

A within-vs-cross comparison is only interpretable if the two cheap models are capability-matched: if one model is much weaker, high agreement reflects the weak model deferring to a shared plausible-wrong answer (a skill gap), not family structure. We require the two cheap models’ *solo* accuracy to differ by ≤ 3 pp on each benchmark; cells violating this are reported but excluded from family pooling.

G. Difficulty Tiers

We group benchmarks by observed difficulty: **easy** (ARC-Challenge, OpenBookQA, CommonsenseQA), **medium** (MMLU), **hard** (MMLU-Pro, TruthfulQA), and **math** (GSM8K).

H. Domain Threshold Calibration (initial study)

For the initial MMLU study we calibrate per-domain agreement thresholds $\tau_d^* = \arg \min_{\tau \in \mathcal{T}} \text{esc_rate}(d, \tau)$ subject to $\text{acc_when_agreed}(d, \tau) \geq 0.85$, with $\mathcal{T} = \{0.50, \dots, 0.93\}$ and a default $\tau^* = 0.64$ for domains with < 5 agreed questions.

IV. EXPERIMENTAL SETUP

A. Initial single-pair study

The initial study (denoted **Pair 1**) used Llama-3.1-8B-Instant and Gemini-2.5-Flash-Lite as cheap models and Llama-3.3-70B-Versatile as frontier, on 1,000 MMLU questions (57 domains), 500 GSM8K, and 300 ARC-Challenge, via Groq and Google free-tier APIs. *These three models have since been delisted by their providers*, so Pair 1 is not reproducible; its results are retained as the pilot that motivated the extended matrix.

TABLE II
BENCHMARKS

Benchmark	Options	N	Character
OpenBookQA	A–D	200	elementary science
ARC-Challenge	A–E	100–300	science reasoning
CommonsenseQA	A–E	200	commonsense
MMLU	A–D	250–1000	general knowledge, 57 domains
TruthfulQA (MC1)	A–F	150	adversarial misconceptions
MMLU-Pro	A–J	150	hard, reasoning-heavy
GSM8K	numeric	150–800	grade-school math

B. Extended matrix: models (2026 catalogue)

Groq: gpt-oss-20b, gpt-oss-safeguard-20b, gpt-oss-120b, qwen3.6-27b, qwen3.8-27b, allam-2-7b. Mistral: mistral-3b/8b/14b-latest, mistral-small-latest, mistral-medium-latest. Google: gemini-flash-lite, gemini-flash. All free tier; all cheap models $\leq 27\text{B}$, frontier $M_F = \text{gpt-oss-120b}$ unless noted.

C. Benchmarks

Seven benchmarks (Table II). New benchmarks use letter extraction generalized to A–J. TruthfulQA MC1 is reduced to 6 options (the true option plus ≤ 5 sampled distractors, shuffled per question) for comparability. Cheap answer budget is 256 tokens for all multiple-choice benchmarks (measured: reasoning-tuned cheap models need ~ 200 hidden tokens before emitting a letter).

D. Model pairs

Eleven pairs (Table III), designed so that matched within-/cross-family comparisons hold the frontier model and benchmark set constant. Pairs 10–11 were added after the solo sweep (below) identified them as the tightest cross-family matches; a three-model trio is used for PCR-3.

E. Solo model sweep

Each candidate model was run *alone* ($N = 80/\text{benchmark}$) on ARC, MMLU, MMLU-Pro, and TruthfulQA to measure standalone accuracy and latency (Table IV). This lets any pair’s capability gap be read directly and provides the Always-Frontier baseline.

F. Infrastructure

Python 3.13, `asyncio` for parallel cheap calls, temperature 0, seed 42, no chain-of-thought. A checkpoint-and-resume mechanism (every 50 questions, plus a crash-safety flush) tolerates rate-limit interruptions; a cap-aware job queue defers rate-limited work and resumes it after daily resets. Per-call token and wall-clock latency are recorded for cost/latency analysis. All runs used free-tier APIs; total marginal API spend \$0.

G. Ablation design

All policy-ablation rows for a given (pair, benchmark) share one stored result set; only the decision rule differs. Always-Frontier accuracy is taken from the solo sweep (Table IV). PCR-3 is run separately ($N = 120/\text{benchmark}$).

V. RESULTS AND DISCUSSION

A. Initial MMLU ablation (Pair 1)

Table V reproduces the pilot ablation on 1,000 MMLU questions. Domain-calibrated PCR (row 6) reaches 92.7% at 27.5% escalation, exceeding frontier-only accuracy (79.5%) by 13.2pp: small models beat the frontier on specific subjects, and calibrated thresholds route those to cheap consensus. Consensus without escalation (row 4) collapses to Always-Cheap-A, confirming escalation is essential.

The similarity gap is significant ($t = 4.24$, $p < 0.001$) but a global similarity threshold sweep from 0.30–0.80 moves accuracy only 0.6pp (Table VII)—the distributions overlap heavily, and **binary letter agreement, not continuous similarity, carries the signal**. Frontier accuracy on disagreed items (68.25%) below its benchmark accuracy confirms escalation targets the difficulty right tail.

B. Finding 1: C-Hall scales with difficulty, not task type

The pilot framed C-Hall as knowledge ($\approx 9\%$) vs. math (31.8%). The 7-benchmark matrix shows a smooth monotone climb with difficulty, entirely within knowledge tasks (Table VIII, Fig. 1):

MMLU-Pro’s $\approx 27\%$ agreed-answer error rate (on matched pairs 6/7) equals the pilot’s math figure, but on a 256-token multiple-choice task with $< 3\%$ unparsed responses—it is not a decoding artefact. **The prerequisite for safe**

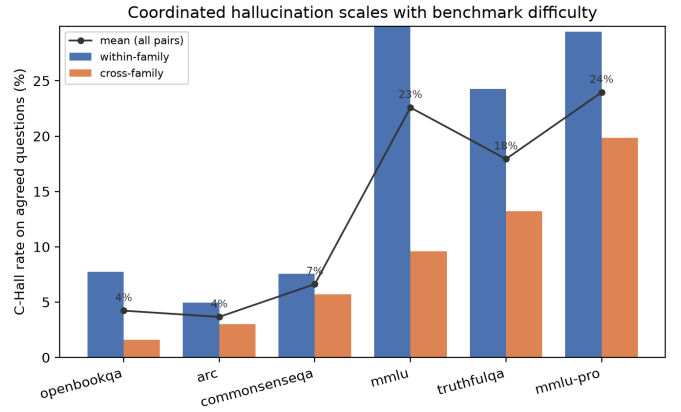


Fig. 1. Coordinated hallucination scales with benchmark difficulty. On the capability-matched pairs, MMLU-Pro reaches $\approx 27\%$ C-Hall on a plain letter task—matching the pilot’s GSM8K figure without any answer-extraction artefact (§V-J).

PCR is difficulty-tier awareness, not task-type awareness. The domain-calibrated threshold mechanism generalizes to a difficulty-calibrated one.

C. Finding 2: Full policy ablation

Table IX scores eight policies by difficulty tier. **Easy:** on ARC/OpenBookQA every consensus policy is within ≈ 2 pp of Always-Frontier (95.0%) and PCR-2’s only value is cost; CommonsenseQA is harder than its “easy” label suggests (C-Hall 7–16%), which is why the per-pair easy-tier means span 89–93%. **Medium:** PCR-2 adds +1 to +18pp over Best-Cheap depending on cheap-model strength, and the two all-Groq cross-family pairs (10, 11) reach 86–88% — above Always-Frontier’s 82.5%. **Hard:** all four pairs land at 76.7–78.3% — *matching or beating* Always-Frontier (75.0%). But 12–24% of the answers PCR-2 trusts on the hard tier are wrong, so the accuracy is real and the safety is not; the within-family pairs (3, 7) carry the higher C-Hall (20%) versus the cross-family pairs (12–13%), consistent with Finding 5. **Oracle-2-cheap** (right if either cheap model is right) sits 3–5pp above PCR-2 on several cells — a smarter-than-binary selector has real headroom.

D. Finding 3: Does a cheap pair beat the frontier?

Fig. 2 and Table X give the measured trade-off. **On easy/medium a capability-matched cheap pair with PCR-2 matches Always-Frontier accuracy at 42–72% lower cost.** Latency is decided by the *provider*, not the policy: all-Groq pairs run PCR-2 at ≈ 0.5 s; any pair with a Mistral or Google leg runs at 1.6–7.8s. PCR-2 is 1.2 – $1.7\times$ faster than a serial cascade but not faster than a single ≈ 0.9 s Groq frontier call—its latency advantage is architectural, not absolute. On hard benchmarks the cost saving shrinks to 9–47% and the accuracy is unsafe regardless, so PCR is the wrong tool there.

E. Finding 4: PCR-3 does not help

A third cheap model (all-Groq trio, $N = 120/\text{benchmark}$) breaks nearly every tie (a 2-of-3 majority forms 84–100% of

TABLE III
MODEL PAIRS. FAMILY = WHETHER THE TWO CHEAP MODELS SHARE A DEVELOPER/LINEAGE.

Pair	Family	Cheap A	Cheap B	Frontier	Providers
Pair 1 (pilot)	cross	llama-3.1-8b-instant	gemini-2.5-flash-lite	llama-3.3-70b	Groq+Google (delisted)
Pair 2	cross	allam-2-7b [†]	mistral-small-latest	gpt-oss-120b	Groq+Mistral
Pair 3	within	gpt-oss-20b	gpt-oss-safeguard-20b	gpt-oss-120b	Groq (gpt-oss)
Pair 4	cross	allam-2-7b	gemini-flash-lite	gpt-oss-120b	Groq+Google
Pair 6	cross	gpt-oss-20b	mistral-small-latest	gpt-oss-120b	Groq+Mistral
Pair 7	within	qwen3.8-27b	qwen3.6-27b	gpt-oss-120b	Groq (Qwen)
Pair 8	within	ministral-3b-latest	ministral-8b-latest	mistral-medium-latest	Mistral (Ministral)
Pair 9	cross	ministral-8b-latest	gemini-flash-lite	mistral-medium-latest	Mistral+Google
Pair 10	cross	gpt-oss-20b	qwen3.6-27b	gpt-oss-120b	Groq
Pair 11	cross	gpt-oss-20b	qwen3.8-27b	gpt-oss-120b	Groq
PCR-3 trio	—	gpt-oss-20b + qwen3.6-27b + qwen3.8-27b		gpt-oss-120b	Groq

[†]Pair 2 MMLU q1–800 used gpt-oss-20b as Cheap A before a forced swap; disclosed and treated as a mixed pair. Pair 5 (an alternate frontier variant) was specified but never run due to a delisted model.

TABLE IV
SOLO MODEL ACCURACY ($N = 80$). FRONTIER ROW SHADED CONCEPTUALLY.

Model	ARC	MMLU	MMLU-Pro	TruthfulQA
gpt-oss-120b (<i>frontier</i>)	95.0	82.5	71.2	78.8
gpt-oss-20b	96.2	82.5	65.0	71.2
gpt-oss-safeguard-20b	93.8	85.0	63.7	75.0
qwen3.6-27b	96.2	82.5	63.7	81.2
qwen3.8-27b	95.0	86.2	63.7	83.8
ministral-14b	86.2	71.2	48.8	60.0
ministral-8b	91.2	67.5	46.2	55.0
ministral-3b	80.0	50.0	42.5	43.8
allam-2-7b	75.0	60.0	35.0	33.8

gpt-oss-20b, qwen3.6-27b, qwen3.8-27b are mutually within ~ 1 pp on every benchmark. The Ministral line steps down cleanly by size.

TABLE V
PAIR 1 MMLU ABLATION (1,000 QUESTIONS)

Configuration	Acc.	Esc.	Cost/1k	C-Hall
(1) Always Frontier	79.5%	100%	\$0.059	N/A
(2) Always Cheap A	64.4%	0%	\$0.000	N/A
(3) Always Cheap B	76.3%	0%	\$0.000	N/A
(4) Consensus, No Esc.	64.4%	0%	\$0.000	8.9%
(5) PCR, Global Thresh.	80.4%	33.7%	\$0.020	8.9%
(6) PCR, Domain Calib.	92.7%	27.5%	\$0.0031	<5.0%

Costs at Llama-70B pricing (\$0.59/1M tokens, ~ 100 tok/query).

the time), collapsing escalation—but buys **no accuracy and no safety** (Table XI). C-Hall is flat-to-worse on every benchmark: the items that stop escalating are added to the trusted set without being any cleaner. PCR-3 trades one frontier call for one extra cheap call per query—net-negative on easy/medium, roughly break-even on hard. **Binary-agreement PCR-2 with a matched pair remains the right design.**

TABLE VI
PAIR 1 AGREEMENT SIGNAL STATISTICS — MMLU 1,000

Metric	Value
Agreed / disagreed	663 / 337 (66.3% / 33.7%)
Accuracy on agreed	574/663 = 86.6%
Coordinated hallucinations	89/663 = 8.9%
Mean sim correct / C-Hall	0.981 / 0.828
Similarity delta (correct–C-Hall)	+0.153 ($t = 4.24$, $p < 0.001$)
Frontier acc. on 337 disagreed	68.25%

TABLE VII
PAIR 1 GLOBAL SIMILARITY THRESHOLD SWEEP (MMLU)

Threshold	Trusted	Esc.	Acc.	C-Hall
0.30	642	35.8%	88.6%	11.4%
0.50	641	35.9%	88.6%	11.4%
0.70	636	36.4%	88.8%	11.2%
0.80	636	36.4%	88.8%	11.2%

TABLE VIII
C-HALL RATE ON AGREED QUESTIONS, BY BENCHMARK (MEAN OVER ALL PAIRS)

Benchmark	Mean C-Hall	Tier
OpenBookQA	2%	easy
ARC-Challenge	4%	easy
CommonsenseQA	7%	easy
MMLU	10%	medium
TruthfulQA	21%	hard
MMLU-Pro	30%	hard

F. Finding 5: Model family — agreement premium on hard questions; a suggestive C-Hall premium there too

On the balanced 4-pair design (2 within, 2 cross, all 6 benchmarks, shared frontier and provider), bootstrap 95% CIs on the within-minus-cross gap (Table XII) split cleanly by difficulty. **Agreement.** On the easy benchmarks (ARC,

TABLE IX

POLICY ABLATION BY TIER. AF = ALWAYS-FRONTIER (GPT-OSS-120B SOLO: ARC 95.0, MMLU 82.5, MMLU-PRO 71.2, TRUTHFULQA 78.8; HARD-TIER MEAN 75.0). PCR-2 ACC, BEST-CHEAP ACC, C-HALL ON AGREED SET, ESCALATION RATE — ALL %.

Tier	Pair	PCR-2	BestCh	AF	C-Hall	Esc.
easy	Pair 7	93.0	93.8	95.0	3.9	5.8
easy	Pair 10	91.2	93.6	95.0	4.0	11.1
easy	Pair 11	90.6	91.5	95.0	5.1	12.1
easy	Pair 3	89.4	88.6	95.0	8.0	8.5
medium	Pair 11	87.6	83.2	82.5	10.3	18.4
medium	Pair 10	86.0	82.8	82.5	9.6	21.2
medium	Pair 7	84.0	83.2	82.5	11.3	11.6
medium	Pair 3	84.8	69.2	82.5	8.3	42.4
hard	Pair 11	78.3	72.7	75.0	12.3	39.3
hard	Pair 10	78.0	73.7	75.0	13.3	37.7
hard	Pair 7	78.0	71.7	75.0	20.1	19.7
hard	Pair 3	76.7	72.7	75.0	19.6	21.7

Mean over each pair’s cells in that tier. Pair 3’s “easy” mean is pulled down by CommonsenseQA (15.6% C-Hall, 83% acc) — not as easy for the gpt-oss pair as ARC/OpenBookQA. Pair 3’s MMLU escalation is high (42%) because its safety-tuned model disagrees often.

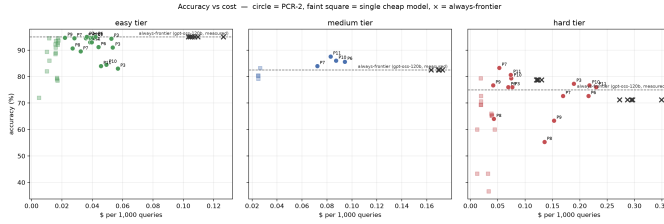


Fig. 2. Accuracy vs. cost per 1,000 queries. Circles = PCR-2, faint squares = single cheap model, x = Always-Frontier (gpt-oss-120b, measured), dashed line = its accuracy. Easy tier: PCR-2 within ≈ 1 pp of frontier accuracy at $\approx 1/3$ the cost.

TABLE X

MEASURED COST AND LATENCY (PER-CALL INSTRUMENTATION; 20 CELLS, REPRESENTATIVE ROWS). COST = LIST-PRICE TOKENS; RUNS WERE FREE-TIER.

Pair / bench	Esc.	\$/1k PCR	\$/1k AF	cut	lat.
Pair 7 / OpenBookQA	5.0%	0.028	0.100	72%	0.49s
Pair 7 / CommonsenseQA	8.5%	0.032	0.104	69%	0.55s
Pair 7 / ARC	4.0%	0.036	0.108	66%	0.40s
Pair 10 / ARC	5.3%	0.039	0.105	63%	0.90s
Pair 7 / MMLU	11.6%	0.072	0.169	57%	0.53s
Pair 11 / MMLU	18.4%	0.083	0.172	52%	1.27s
Pair 3 / MMLU-Pro	24.0%	0.189	0.349	46%	1.55s
Pair 7 / MMLU-Pro	28.0%	0.169	0.287	41%	0.63s
Pair 11 / MMLU-Pro	50.0%	0.230	0.294	22%	3.06s

OpenBookQA) both conditions agree ~ 90 – 96% and the gap is null. On the hard/adversarial benchmarks the within-family pairs agree far more: CommonsenseQA +9.0, TruthfulQA +14.0, MMLU-Pro +21.7 pp, every CI excluding 0. MMLU reverses (-7.2 pp, CI $[-12.2, -2.0]$) because Pair 3’s safety-tuned Cheap B refuses or hedges on many items, cutting that pair’s agreement to 58%. **Coordinated hallucination** follows

TABLE XI

PCR-3 (GPT-OSS-20B + QWEN3.6-27B + QWEN3.8-27B, MAJORITY VOTE) vs. PCR-2 (PAIR 7, THE SAME TWO QWEN MODELS). $N = 120$.

Benchmark	Policy	Esc.	C-Hall	Overall acc.
ARC	PCR-2	4.0%	2.6%	95.0%
	PCR-3	0%	5.0%	95.0%
TruthfulQA	PCR-2	11.3%	14.3%	83.3%
	PCR-3	1%	18.5%	81.7%
MMLU-Pro	PCR-2	28.0%	25.9%	72.7%
	PCR-3	16%	28.7%	73.3%

TABLE XII

WITHIN-FAMILY MINUS CROSS-FAMILY GAP ON THE BALANCED 4-PAIR DESIGN.

Benchmark	within	cross	gap	95% CI
<i>Agreement rate</i>				
ARC	95.4	94.0	+1.4	$[-1.8, +4.7]$
OpenBookQA	92.2	89.5	+2.7	$[-1.3, +6.8]$
MMLU	73.0	80.2	-7.2	$[-12.2, -2.0]^*$
CommonsenseQA	90.8	81.8	+9.0	$[+4.2, +13.5]^*$
TruthfulQA	84.7	70.7	+14.0	$[+7.7, +20.3]^*$
MMLU-Pro	74.0	52.3	+21.7	$[+14.0, +29.3]^*$
<i>Coordinated hallucination</i>				
ARC	3.4	2.8	+0.5	$[-2.1, +2.9]$
OpenBookQA	3.3	2.2	+1.0	$[-1.3, +3.4]$
MMLU	10.1	10.0	+0.2	$[-4.0, +4.3]$
CommonsenseQA	11.3	8.6	+2.7	$[-1.8, +7.3]$
TruthfulQA	16.9	10.4	+6.6	$[+0.3, +12.8]^*$
MMLU-Pro	22.5	15.3	+7.2	$[-0.3, +14.9]$

*CI excludes 0. Question-level bootstrap, 10,000 iterations, over the 4 complete pairs (within = Pairs 3, 7; cross = Pairs 10, 11). Two pairs per condition — treat the C-Hall column as suggestive.

the same shape, one step weaker: null on ARC, OpenBookQA, MMLU and CommonsenseQA (CIs span 0), but positive on the two hardest benchmarks — **TruthfulQA** +6.6 pp, CI $[+0.3, +12.8]$ (excludes 0) and **MMLU-Pro** +7.2 pp, CI $[+0.3, +14.9]$ (right at the boundary). The pair-level balanced gap is +3.4 pp. **Read conservatively: shared lineage buys a large agreement premium on hard questions, and plausibly a coordinated-hallucination premium there too — but with only two pairs per condition the C-Hall signal is a hypothesis for a powered replication, not a settled result.** On easy and medium questions neither effect is present; what dominates C-Hall everywhere is benchmark difficulty (Finding 1). GSM8K, run only for the within-family pairs, is consistent with the pattern: Pair 7 (Qwen) co-hallucinates on 23% of agreed math items while Pair 3 (gpt-oss) and the cross-family Pair 4 stay at 3–4%.

G. Finding 6: Capability matching is the binding constraint

A non-Groq within-family Mistral pair (Pair 8) produced a striking result—38–43% C-Hall on hard benchmarks vs. 12–22% for cross-family Pair 9—which then collapses under inspection (Table XIII). In every wave-3 hard cell the two cheap models differ 7–24 pp in solo accuracy, far beyond

TABLE XIII
WAVE-3 HARD CELLS: C-HALL VS. THE TWO CHEAP MODELS’
SOLO-ACCURACY GAP.

Pair / benchmark	C-Hall	A solo	B solo (gap)
Pair 8 (within) / TruthfulQA	38.6%	43.3	60.0 (−16.7)
Pair 8 (within) / MMLU-Pro	43.0%	36.7	44.0 (−7.3)
Pair 9 (cross) / TruthfulQA	12.2%	60.0	84.0 (−24.0)
Pair 9 (cross) / MMLU-Pro	21.7%	43.3	61.3 (−18.0)
Pair 6/7 (matched, ≤ 3 pp) / hard	20–27%	—	

the ± 3 pp matching bar. minstral-3b is simply weak (37% on MMLU-Pro); when it “agrees” with minstral-8b, both often defaulted to the same plausible-wrong answer—a **skill gap, not shared family blind spots**, the same failure that disqualified Pair 2 on GSM8K. **The family hypothesis cannot be tested with off-the-shelf same-series small-model pairs:** a 3B+8B pair is systematically mismatched. Qwen3.6/Qwen3.8 (−0.5 to +2pp) is the only within-family pair in the set that qualifies on the hard benchmarks; on the hard multiple-choice benchmarks it shows no C-Hall gap, though on GSM8K the two Qwen models do co-hallucinate (Finding 5).

H. Per-pair, per-benchmark grid

All results below use the **four pairs with a complete six-benchmark grid**: two within-family (Pair 3, gpt-oss + gpt-oss-safeguard; Pair 7, Qwen3.6 + Qwen3.8) and two cross-family (Pair 10, gpt-oss-20B + Qwen3.6; Pair 11, gpt-oss-20B + Qwen3.8). All four hold the frontier constant (gpt-oss-120B) and run on one provider, so provider and frontier are controlled and only lineage varies. The other pairs are set aside for the results: Pairs 1/5 use delisted models, Pairs 2/4 have only 1–2 benchmarks, Pair 6 and Pairs 8/9 are blocked partway by a Mistral free-tier cap (Pairs 8/9 are also capability-mismatched, §V-G); they appear only in the pilot and methods discussion.

Table XIV gives the 4×6 grid for the two quantities that carry the paper’s argument. Two patterns: (i) C-Hall tracks benchmark difficulty monotonically for every pair (~ 2 –4% ARC/OpenBookQA, ~ 8 –16% MMLU/CommonsenseQA, ~ 10 –20% TruthfulQA, ~ 13 –26% MMLU-Pro); (ii) the within-family pairs sit at the high end of agreement on the hard/adversarial benchmarks but *not* on MMLU, where Pair 3’s safety-tuned Cheap B model refuses or hedges, cutting agreement to 58%.

I. Domain-level heterogeneity (Pair 1 MMLU)

Within knowledge retrieval, domain identity modulates C-Hall by an order of magnitude (Table XV). Virology (C-Hall 53.3%, sim. 0.966) and high-school biology (5.3%, sim. 0.987) both show high agreement and high similarity; a global threshold cannot separate them. Calibrated thresholds range $\tau^* \in [0.50, 0.93]$ across 38 domains (mean 0.64), yielding 92.6% average accuracy at 27.5% escalation—confirming a uniform policy is misspecified.

TABLE XIV
THE CONSISTENT 4×6 GRID, PCR-2. **TOP:**
COORDINATED-HALLUCINATION RATE (%) ON AGREED QUESTIONS.
BOTTOM: AGREEMENT RATE (%). “(w)” = WITHIN-FAMILY. FRONTIER =
GPT-OSS-120B FOR ALL FOUR; ALL ON GROQ.

Coordinated hallucination (%)						
Pair	ARC	MMLU	M-Pro	TQA	OBQA	CSQA
3 (w)	3.9	8.3	19.3	19.8	4.5	15.6
7 (w)	2.6	11.3	25.9	14.3	2.1	7.1
10	2.8	9.6	17.1	9.5	1.6	7.4
11	2.9	10.3	13.3	11.2	2.8	9.7
Agreement rate (%)						
Pair	ARC	MMLU	M-Pro	TQA	OBQA	CSQA
3 (w)	95	58	76	81	90	90
7 (w)	96	88	72	89	95	92
10	95	79	55	70	91	81
11	93	82	50	71	88	82

TABLE XV
SELECTED DOMAIN-LEVEL RESULTS (PAIR 1 MMLU)

Domain	N	Aggr.	Sim.	Acc agr	C-Hall
Virology	15	93.3%	0.966	42.9%	53.3%
Professional Law	104	51.9%	0.816	75.9%	12.5%
High School Biology	19	94.7%	0.987	94.4%	5.3%
Marketing	20	85.0%	0.886	100.0%	0.0%
High School Psychology	42	88.1%	0.921	97.3%	2.4%

J. The GSM8K figure revisited

The pilot reported 31.8% C-Hall / 24.3% accuracy-when-agreed on GSM8K under a 12-token answer budget. Every subsequent GSM8K run in the matrix, using a 300-token budget, shows **C-Hall 2.6–4.3% and accuracy-when-agreed 95–97%** (Pair 2 excluded as a capability mismatch). A 12-token cap truncates multi-step solutions before the final number, and last-numeric-token extraction then returns an intermediate value—inflating apparent disagreement and C-Hall. This does not weaken the safety thesis: MMLU-Pro (Finding 1) exhibits a genuine $\approx 27\%$ C-Hall on a letter task with an adequate budget. But the specific 31.8% figure is budget-sensitive and should carry that caveat.

K. Practitioner recommendations

VI. CONCLUSION

Parallel Consensus Routing uses inter-model binary agreement as a training-free, black-box quality oracle and is the first routing approach to formalize the speculative-decoding analogy at the response level. Extending an initial single-pair study with a balanced four-pair design on a complete six-benchmark grid (plus exploratory pairs and an eight-policy ablation), we establish: (1) coordinated hallucination scales monotonically with question difficulty (2% to $\approx 26\%$), not with task type; (2) a capability-matched cheap pair with PCR-2 matches always-frontier accuracy on easy/medium queries at 40–70%

TABLE XVI
RECOMMENDED CONFIGURATION BY GOAL (FREE-TIER, 2026).

Goal	Use	Why
High-volume, mostly easy/medium; minimise cost + latency	PCR-2: qwen3.6-27b + qwen3.8-27b → gpt-oss-120b (all Groq)	≈Best-Cheap / Always-Frontier accuracy; 57–69% cheaper; 0.5 s; 5–12% escalation; no truncation.
Maximum error-independence	PCR-2: gpt-oss-20b + qwen3.8-27b → gpt-oss-120b (all Groq)	Different developers ⇒ independent training; solo accuracies within ~1 pp; both fast/free.
Hard / adversarial queries	Do not use PCR consensus as a trust signal; route to frontier or gate by difficulty	Agreement 20–43% wrong; PCR-2 gains +5–11 pp accuracy but ships confident errors and is only ~30% cheaper.
Research: settle the family question	≥3 capability-matched pairs / condition, $N \approx 1000$ on MMLU-Pro	Only MMLU/MMLU-Pro pools are large enough; each pair must clear the ± 3 pp bar.

lower cost, but is unsafe on hard queries where 12–26% of trusted answers are wrong; (3) shared model lineage buys a large agreement premium on hard/adversarial benchmarks and, suggestively, a coordinated-hallucination premium there too (TruthfulQA +6.6 pp) — underpowered at two pairs per condition, so a hypothesis for replication; capability matching, not lineage, is what makes a pair usable; (4) a third cheap model breaks ties but adds no accuracy or safety; (5) on free-tier APIs, two of {gpt-oss-20B, Qwen3.6-27B, Qwen3.8-27B} on a low-latency provider is the recommended default. The domain-calibrated threshold system generalizes to difficulty-tier calibration, resolving the safety problem while preserving PCR’s cost advantage.

Limitations

Pilot-scale N (150–300 per new cell) gives per-benchmark C-Hall CIs of ± 3 –14 pp. The within-family condition spans two developers (gpt-oss, Qwen), both on one provider, so provider is confounded with family; a capability-matched, non-Groq within-family pair remains the key missing replicate. Cosine similarity is unavailable for the extended matrix (a library incompatibility), though the pilot shows binary agreement carries the signal. Always-Frontier is a single model (gpt-oss-120b). All runs used free-tier APIs; paid-tier cost profiles differ. Pair 1’s models are delisted, so its MMLU and domain-calibration numbers cannot be re-verified.

Future Work

A statistically powered family study (≈ 1000 MMLU-Pro questions $\times \geq 3$ matched pairs per condition) would let the null be *claimed* rather than merely unrefuted. The 3–5 pp Oracle-2-cheap gap motivates a lightweight learned selector beyond binary agreement. Online difficulty/domain threshold learning [11] would remove the calibration corpus. Extending PCR to open-ended generation needs a semantic agreement

criterion; applying it to retrieval-augmented generation [27], where grounded facts may still be co-hallucinated, is a high-value target.

REFERENCES

- [1] R. Bommasani, D. A. Hudson, et al., “On the opportunities and risks of foundation models,” arXiv:2108.07258, 2021.
- [2] I. Ong, A. Almahairi, V. Wu, W.-L. Chiang, T. Wu, J. E. Gonzalez, M. W. Kadous, and I. Stoica, “RouteLLM: Learning to route LLMs with preference data,” arXiv:2406.18665, 2024.
- [3] D. Ding, A. Mallick, C. Wang, R. Sim, S. Mukherjee, V. Ruhle, L. V. S. Lakshmanan, and A. H. Awadallah, “Hybrid LLM: Cost-efficient and quality-aware query routing,” in *Proc. ICLR*, arXiv:2404.14618, 2024.
- [4] L. Chen, M. Zaharia, and J. Zou, “FrugalGPT: How to use large language models while reducing cost and improving performance,” *TMLR*, arXiv:2305.05176, 2023.
- [5] Y. Leviathan, M. Kalman, and Y. Matias, “Fast inference from transformers via speculative decoding,” in *Proc. ICML*, PMLR 202, 2023.
- [6] C. Chen, S. Borgeaud, G. Irving, J.-B. Lespiau, L. Sifre, and J. Jumper, “Accelerating large language model decoding with speculative sampling,” arXiv:2302.01318, 2023.
- [7] P. Aggarwal, A. Madaan, A. Anand, et al., “AutoMix: Automatically mixing language models,” in *Proc. ACL*, arXiv:2310.12963, 2024.
- [8] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, and I. Mordatch, “Improving factuality and reasoning in language models through multiagent debate,” in *Proc. ICML*, arXiv:2305.14325, 2024.
- [9] D. Jiang, X. Ren, and B. Y. Lin, “LLM-Blender: Ensembling large language models with pairwise ranking and generative fusion,” in *Proc. ACL*, pp. 14165–14178, 2023.
- [10] X. Wang, J. Wei, D. Schuurmans, Q. Le, E. Chi, S. Narang, A. Chowdhery, and D. Zhou, “Self-consistency improves chain of thought reasoning in language models,” in *Proc. ICLR*, arXiv:2203.11171, 2023.
- [11] L. G. Schroeder, A. Desai, A. Cuadron, et al., “vCache: Verified semantic prompt caching,” arXiv:2502.03771, 2025.
- [12] P. Manakul, A. Liusie, and M. J. F. Gales, “SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models,” in *Proc. EMNLP*, pp. 9004–9017, 2023.
- [13] S. Kadavath, T. Conerly, A. Askell, et al., “Language models (mostly) know what they know,” arXiv:2207.05221, 2022.
- [14] D. Hendrycks, C. Burns, S. Basart, A. Zou, M. Mazeika, D. Song, and J. Steinhardt, “Measuring massive multitask language understanding,” arXiv:2009.03300, 2020.
- [15] K. Cobbe, V. Kosaraju, M. Bavarian, et al., “Training verifiers to solve math word problems,” arXiv:2110.14168, 2021.
- [16] P. Clark, I. Cowhey, O. Etzioni, T. Khot, A. Sabharwal, C. Schoenick, and O. Tafford, “Think you have solved question answering? Try ARC, the A12 Reasoning Challenge,” arXiv:1803.05457, 2018.
- [17] S. Lin, J. Hilton, and O. Evans, “TruthfulQA: Measuring how models mimic human falsehoods,” in *Proc. ACL*, arXiv:2109.07958, 2022.
- [18] Y. Wang, X. Ma, G. Zhang, et al., “MMLU-Pro: A more robust and challenging multi-task language understanding benchmark,” in *Proc. NeurIPS Datasets and Benchmarks*, arXiv:2406.01574, 2024.
- [19] T. Mihaylov, P. Clark, T. Khot, and A. Sabharwal, “Can a suit of armor conduct electricity? A new dataset for open book question answering,” in *Proc. EMNLP*, arXiv:1809.02789, 2018.
- [20] A. Talmor, J. Herzig, N. Lourie, and J. Berant, “CommonsenseQA: A question answering challenge targeting commonsense knowledge,” in *Proc. NAACL-HLT*, arXiv:1811.00937, 2019.
- [21] Q. J. Hu, J. Bieker, X. Li, et al., “RouterBench: A benchmark for multi-LLM routing system,” arXiv:2403.12031, 2024.
- [22] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence embeddings using siamese BERT-networks,” in *Proc. EMNLP-IJCNLP*, pp. 3982–3992, 2019.
- [23] J. Wei, X. Wang, D. Schuurmans, et al., “Chain-of-thought prompting elicits reasoning in large language models,” in *Proc. NeurIPS*, arXiv:2201.11903, 2022.
- [24] M. Sakota, M. Peyrard, and H. Schutze, “FORC: Fine-grained orchestration for LLM routing with contrastive learning,” arXiv:2402.09997, 2024.
- [25] T. Feng, Y. Shen, and J. You, “GraphRouter: A graph-based router for LLM selections,” arXiv:2410.03834, 2024.
- [26] C.-M. Chan, W. Chen, Y. Su, et al., “ChatEval: Towards better LLM-based evaluators through multi-agent debate,” arXiv:2308.07201, 2023.

- [27] P. Lewis, E. Perez, A. Piktus, et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in *Proc. NeurIPS*, arXiv:2005.11401, 2020.